

HMF Design Clinic Workbook

Cedarline Health Group, a Hybrid Mesh Firewall design mission

Today your team is the Cisco Solution Engineering team walking into Cedarline. You will sit with four real conversations, hear what is keeping them up at night, and answer as Solution Engineers do, with the right solution, why it fits, and the outcome it delivers, all brought together into one Hybrid Mesh Firewall design.

The scenario. Cedarline Health Group is a regional healthcare payer-provider with roughly 8,000 employees across four hospitals and thirty-one clinics, plus a claims-processing business that handles the money. Growth by acquisition (including a second data centre absorbed as-is) and a mid-flight, disorderly move to public cloud have left a fragmented security posture. After a security incident in March, the board asked what would have stopped it, and leadership still cannot answer. Budget and executive attention are approved but explicitly temporary, and the security team is small.

TEAM

MEMBERS (UP TO 8)

DATE

What you will walk away able to do

- 1 Map Cedarline's stated concerns to the right Cisco Secure solutions.
- 2 Explain, in the customer's language, how each solution addresses the concern.
- 3 Connect the four case studies into one coherent Hybrid Mesh Firewall design.
- 4 Present the business outcome, not just the feature.

How your answers are assessed

Each case study is scored across its four pain points, giving a case study score and an overall score for the clinic. Alongside the score, your answers are assessed on three dimensions.

Problem Identifier

Correctly identified the customer's pain point and the requirements behind it.

Solution Designer

Recommended the right Cisco solution, architecture, and approach.

Value Communicator

Clearly explained how the solution works and the customer outcomes it enables.

The two-part answer. For every customer concern you will write a **Product / platform** and a **Solution Engineer response** that covers how it addresses the concern and the outcome it delivers. Identify the problem, design the right solution, and communicate the value.

Before you submit, self-check

- Problem: did we name the real requirement behind the concern, not just the surface ask?
- Solution: did we recommend a specific Cisco product and the approach, not only a product name?
- Value: did we explain how it works and the outcome for Cedarline?

What a strong answer looks like

This is an example only, from a different customer conversation. The four case studies you will work are different. Notice how the response identifies the problem, designs the solution, and communicates the value.

Our support team uses a public AI chatbot to draft replies, and staff sometimes paste patient details into it. We cannot see or control what leaves the organisation.

PRODUCT / PLATFORM

Cisco AI Defense

SOLUTION ENGINEER RESPONSE

Identify. The real requirement is preventing sensitive patient data from leaving the organisation through an unsanctioned AI app, with visibility and control over how it is used. **Design.** Cisco AI Defense discovers the AI applications in use and applies runtime guardrails to the prompt and response traffic, so risky data is caught in real time. **Communicate.** This lets the support team keep using AI safely rather than banning it, protects patient data and compliance, and adds no friction to their day-to-day work.

Product glossary

Every Cisco product, capability and term used in the clinic. Which product fits which concern is for you to decide.

PLATFORMS AND PRODUCTS

Cisco AI Defense	Secures AI applications and models across build and runtime. It validates models and applications, scans the AI supply chain (models, repositories and MCP servers), discovers AI assets across the cloud, and applies runtime guardrails to prompt and response traffic.
Cisco Multicloud Defense	One security policy plane across public clouds. It provides consistent segmentation, egress control, traffic visibility and policy enforcement across AWS, Azure and GCP, and integrates with AI Defense for cloud runtime enforcement.
Cisco Secure Firewall (FTD)	Cisco's next-generation firewall, Firewall Threat Defense. It delivers perimeter and data-centre enforcement with intrusion prevention, application control and advanced threat detection.
Cisco Secure Firewall Management Center (FMC)	The centralised manager for the Secure Firewall estate, handling policy, logging and reporting across firewalls.
Cisco Security Cloud Control	Cloud-delivered management across the firewall and security estate, for unified policy administration and rule cleanup at scale.
Cisco Secure Workload	Application-dependency mapping and micro-segmentation. It reveals how workloads actually communicate, then drives safe segmentation policy before it is enforced.
Cisco Identity Services Engine (ISE)	Network identity and access control. It knows who and what is on the network and enables rapid, automated quarantine of compromised devices.
Cisco Hypershield	AI-native, highly distributed security enforcement. It provides autonomous, in-kernel (eBPF) segmentation and protection for workloads at scale.
Isovalent Cilium	eBPF-based Kubernetes networking and policy, giving consistent connectivity and policy enforcement across clusters and environments.
Isovalent Tetragon	eBPF-based runtime security for workloads, detecting suspicious system and network behaviour in real time.

KEY CAPABILITIES AND FEATURES

Encrypted Visibility Engine (EVE)	A Secure Firewall capability that infers threats from encrypted traffic patterns without decrypting the payload.
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Snort ML	A Secure Firewall capability that adds machine-learning detection inside the Snort 3 engine to catch unknown and zero-day exploits.
pxGrid	An ISE capability: a context-sharing fabric so security tools can act on identity, for example triggering an automatic quarantine.
Adaptive Network Control (ANC)	An ISE capability that quarantines or restricts a device in real time when a threat is detected.
Security Group Tags (SGT)	Group-based policy tags (Cisco TrustSec) that enable scalable, identity-aware segmentation.

ARCHITECTURE

Hybrid Mesh Firewall (HMF)	Cisco's approach to consistent security enforcement across firewalls, cloud gateways, workloads and Kubernetes, all managed centrally.
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Decode the jargon

EU AI Act, Article 15	EU law classifying AI by risk. A 'Provider' must prove robustness and cybersecurity.
MCP (Model Context Protocol)	How AI agents connect to tools and data, creating a new supply-chain surface to secure.
East-west traffic	Traffic between workloads inside the data centre, as opposed to north-south, which flows in and out.
Lateral movement	An attacker hopping from one compromised system to others on the inside.
Micro-segmentation	Fine-grained isolation of individual workloads so a breach cannot spread.
eBPF	Linux kernel technology used to observe and enforce networking and security with low overhead.
EVE (Encrypted Visibility Engine)	Infers threats from encrypted traffic patterns without decrypting it.
Snort ML	Machine-learning detection inside the Snort engine for unknown and zero-day exploits.
pxGrid	Cisco's context-sharing fabric so tools can act on identity, for example an automatic quarantine via ISE.
Hybrid Mesh Firewall (HMF)	Consistent security enforcement across firewalls, cloud gateways, workloads and Kubernetes, managed centrally.

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Case Study 1

The conversation. Cedarline is standing up AI across regulated workflows and public cloud faster than security can see it. Help them prove compliance, vet what they adopt, and get visibility and guardrails over AI traffic.

PAIN POINT 1 Case Study 1

Under the EU AI Act, we are classified as a 'Provider' for our internal AI systems. We need to demonstrate robustness and cybersecurity compliance to satisfy Article 15. How does Cisco help us meet these specific requirements?

PRODUCT / PLATFORM

SOLUTION ENGINEER RESPONSE: HOW IT ADDRESSES THE CONCERN AND THE OUTCOME

Assessed on three dimensions. **Problem Identifier**: identify the pain point and its requirements. **Solution Designer**: recommend the right Cisco solution, architecture and approach. **Value Communicator**: explain how it works and the outcome it enables.

We are integrating various open-source models and third-party agents into our internal workflows. How can we ensure these assets aren't introducing vulnerabilities into our environment before they even go live?

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We know AI is showing up across our cloud environments faster than security can track it. We're worried we have models, agents, datasets, and MCP-connected tools running without visibility or consistent controls.

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We're running AI workloads in cloud accounts, but we don't have a reliable way to see which models and AI assets are actually in use. We're also concerned about workloads reaching out to third-party AI models without security knowing about it. And even when we do know the traffic path, we need a practical way to inspect and enforce guardrails on prompt and response traffic in the cloud without changing every application.

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Case Study 2

The conversation. Workloads sprawl across AWS, Azure and on-prem, each with its own policy model. Contain lateral movement, prove consistent controls to auditors, and give Cedarline one place to define and enforce policy.

PAIN POINT 1 Case Study 2

If a breach occurs in one of our public cloud instances, we are terrified that the attacker will move laterally into our core on-premises systems. How can we contain these threats in a multicloud environment?

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PAIN POINT 2 Case Study 2

Our auditors require proof that our security controls are applied consistently across all our cloud workloads. Gathering this data from different cloud providers is a manual, time-consuming nightmare.

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We have workloads spread across AWS, Azure, and on-premises data centers. Managing security policies individually in each cloud environment is creating massive operational overhead and leading to inconsistent security posture.

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We have workloads on-prem, in cloud, and in mixed environments, and policy consistency is becoming difficult. Every environment feels different, and we're worried about drift.

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Case Study 3

The conversation. A mixed-vendor perimeter, an inherited rulebase and mostly encrypted traffic leave blind spots and a slow response. Restore visibility, speed the response, and modernise detection with the team they have.

PAIN POINT 1 Case Study 3

When our security tools detect a compromised device, our manual response time is too slow. By the time we identify the user and manually isolate the port, the attacker has already moved through the network.

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PAIN POINT 2 Case Study 3

Our firewall environment has become too complex to manage manually.

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Most of our traffic is encrypted, so we've lost visibility into a huge part of our environment.

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We're worried traditional signature-based detection won't catch new or unknown attacks quickly enough.

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Case Study 4

The conversation. A flat data centre and fragmented Kubernetes networking invite east-west movement. Build segmentation Cedarline can adopt safely, with runtime visibility and application-dependency awareness so nothing breaks.

PAIN POINT 1 Case Study 4

Network policy helps, but we also need to know what's happening at runtime inside the workload. We need better detection of suspicious behavior and policy violations.

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PAIN POINT 2 Case Study 4

Our Kubernetes networking feels fragmented across clusters and environments. We're dealing with too many moving parts, inconsistent policies, and too much operational complexity.

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Right now we're stitching together separate tools for networking, security, and observability in Kubernetes, and it's creating too much overhead.

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PAIN POINT 4 Case Study 4

We know we need microsegmentation, but we don't have enough visibility into east-west flows or application dependencies to create policy safely. We don't want to break the application.

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Your Hybrid Mesh Firewall design

Four conversations, one story. Combine your case studies into a single HMF design and paste your team's topology below. Your facilitator collects this for **Subject Matter Expert (SME) review**: it is reviewed, not scored.

Paste your team's topology diagram here

Show how the four case studies connect across Cedarline's on-prem, data-centre and multicloud estate. Submitted for SME review.